

AYUSH SABHASAD

Undergraduate Student at IIIT Hyderabad

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Experience

Discover Bio-Insights | Software Intern

January 2026 – April 2026 | Remote

- Architected a maternal-child **clinical data platform** featuring offline-first sync, comprehensive care workflows, and visit-based read-only locking.
- Engineered a practitioner management system with role-based access and duplicate detection, cutting data conflicts by 95%.

Education

International Institute of Information Technology, Hyderabad

August 2023 – Present

B.Tech in Computer Science and M.S. in Computing and Human Sciences

Hyderabad, India

Relevant Coursework

- Data Structures and Algorithms
- Probability and Statistics
- Linear Algebra
- Operating Systems and Networks
- Design and Analysis of Software
- Introduction to Software Systems
- Machine, Data and Learning
- Data and Applications

Projects

Military Database | *Python, MySQL, RDBMS*

[GitHub](#)

- Designed a 30+ table **relational database** for Military data, applying 1NF-3NF normalization to eliminate redundancy.
- Implemented 10+ core features, including CRUD operations and access control for streamlined interactions.

C-Shell | *C, Operating Systems, Networking*

[GitHub](#)

- Built a **Unix shell** with command parsing, job control, I/O redirection, and TCP/IP socket programming for client-server communication.
- Implemented FCFS and CFS **CPU scheduling algorithms** in the xv6 kernel using priority weighting and virtual runtime tracking.

Multimedia Management Platform | *Flask, PostgreSQL, Python*

[GitHub](#)

- Developed a secure web app with bcrypt hashing, JWT authentication, and role-based access control for dashboards.
- Designed a PostgreSQL database managing users and multimedia, featuring secure file uploads, base64 image galleries, and CSRF protection.

Distributed File Storage Server | *C, POSIX Threads, Socket Programming*

[GitHub](#)

- Developed a multi-threaded distributed storage server using mutex-based file locking for consistent concurrent client-server operations.
- Implemented core operations (READ, WRITE, STREAM, UNDO, DELETE) with automatic backups, atomic modifications, and robust error handling.

Felicity – Campus Event Management Platform | *React.js, Node.js, Express.js, MongoDB*

[GitHub](#)

- Architected a full-stack event platform with Role-Based Access Control (RBAC) for multiple user tiers.
- Engineered an event lifecycle engine to automate state transitions and strictly enforce registration deadlines.

Technical Skills

Languages: C, C++, Python, JavaScript, SQL, HTML, CSS

Web Development Frameworks: Flask, MongoDB, Express, React, Node, Next, React Native

Tools: Git, GitHub, VS Code, Android Studio

Achievements/Extracurricular

- Served as **Team Lead** and **Logistics Member** (SACC, 2023–2026) and **Events Member** (The Dance Crew, 2023–2026) at IIIT Hyderabad, fostering leadership and soft skills.
- Conducted field-based urban studies, engaging with **10+ citizens, officials, and NGO representatives**.
- Developed **qualitative research skills** by interviewing stakeholders and analyzing urban governance challenges.
- Collaborated on on-ground data collection, enhancing **teamwork, critical thinking, and public interaction**.
- Secured **99.61%** in MHT-CET (320K+ applicants) – 2023.